

Rajalakshmi Engineering College

Name: Harivarsha s
Email: 240801109@rajalakshmi.edu.in
Roll no: 240801109
Phone: 6380939059
Branch: REC
Department: I ECE FA
Batch: 2028
Degree: B.E - ECE

Scan to verify results



NeoColab_REC_CS23231_DATA STRUCTURES

REC_DS using C_Week 1_COD_Question 3

Attempt : 1
Total Mark : 10
Marks Obtained : 10

Section 1 : Coding

1. Problem Statement

Imagine you are working on a text processing tool and need to implement a feature that allows users to insert characters at a specific position.

Implement a program that takes user inputs to create a singly linked list of characters and inserts a new character after a given index in the list.

Input Format

The first line of input consists of an integer N, representing the number of characters in the linked list.

The second line consists of a sequence of N characters, representing the linked list.

The third line consists of an integer index, representing the index(0-based) after

which the new character node needs to be inserted.

The fourth line consists of a character value representing the character to be inserted after the given index.

Output Format

If the provided index is out of bounds (larger than the list size):

1. The first line of output prints "Invalid index".
2. The second line prints "Updated list: " followed by the unchanged linked list values.

Otherwise, the output prints "Updated list: " followed by the updated linked list after inserting the new character after the given index.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 5

a b c d e

2

X

Output: Updated list: a b c X d e

Answer

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
// Define the node structure
```

```
struct Node {  
    char data;  
    struct Node* next;  
};
```

```
// Function to create a new node
```

```
struct Node* createNode(char data) {  
    struct Node* newNode = (struct Node*)malloc(sizeof(struct Node));
```

```

    newNode->data = data;
    newNode->next = NULL;
    return newNode;
}

// Function to insert a node after a given index
int insertAfter(struct Node* head, int index, char newChar) {
    struct Node* current = head;
    int count = 0;

    // Traverse to the node before the given index
    while (current != NULL && count < index) {
        current = current->next;
        count++;
    }

    if (current == NULL) { // If index is out of bounds
        return 0; // Indicate failure
    }

    // Create the new node and insert it after the current node
    struct Node* newNode = createNode(newChar);
    newNode->next = current->next;
    current->next = newNode;

    return 1; // Indicate success
}

// Function to print the list
void printList(struct Node* head) {
    struct Node* current = head;
    while (current != NULL) {
        printf("%c ", current->data);
        current = current->next;
    }
    printf("\n");
}

int main() {
    int N, index;
    char newChar;

```

```

// Input the number of characters in the list
scanf("%d", &N);

// Create the head of the linked list and populate it
struct Node* head = NULL;
struct Node* temp = NULL;

for (int i = 0; i < N; i++) {
    char ch;
    scanf(" %c", &ch); // Read each character (space before %c to skip any
                        // leftover newline)

    struct Node* newNode = createNode(ch);
    if (head == NULL) {
        head = newNode; // First node
    } else {
        temp->next = newNode; // Append to the last node
    }
    temp = newNode;
}

// Input the index and the character to insert
scanf("%d", &index);
scanf(" %c", &newChar);

// Try to insert the new character at the given index
if (!insertAfter(head, index, newChar)) {
    printf("Invalid index\n");
}

// Print the updated list
printf("Updated list: ");
printList(head);

return 0;
}

```

Status : Correct

Marks : 10/10